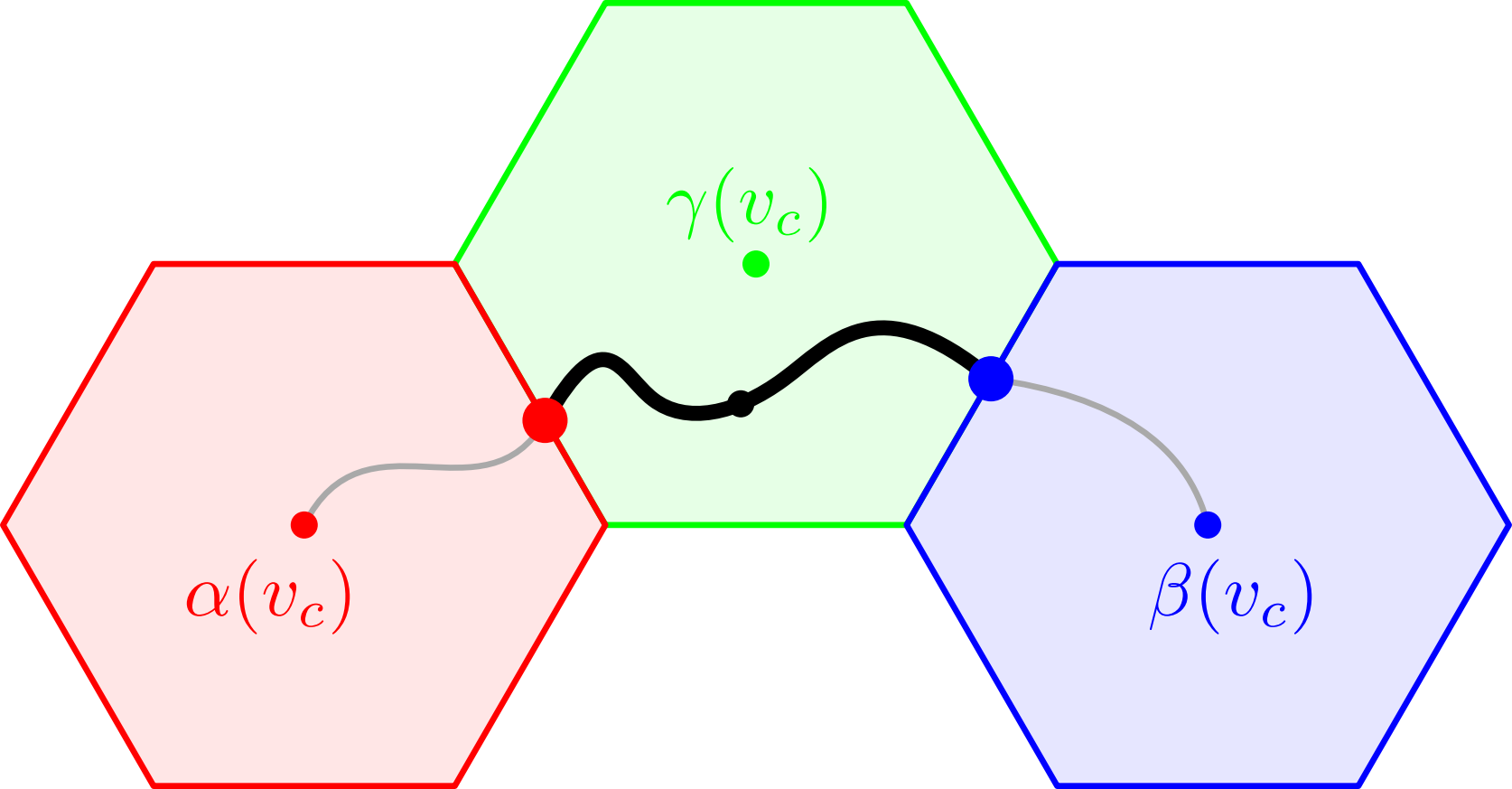
$\gamma(v_c)$
$\alpha(v_c)$
$\beta(v_c)$